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URL: <http://www.neuralengine.org>

Education

Doctor of Philosophy, Kyoto University

2004 Apr - 2007 Mar

Department of Physics, Graduate School of Science, Kyoto University, Kyoto, Japan.

Supervisor: Prof. Shigeru Shinomoto

Thesis title: Recipes for selecting the bin size of a histogram.

Master of Arts (Neuroscience), Johns Hopkins University

2000 Aug - 2003 Nov

Department of Neuroscience, School of Medicine, Baltimore, MD, USA.

Supervisor: Prof. Ernst Niebur

Thesis title: Communication over distributed neural systems.

Bachelor of Engineering, Keio University

1996 Apr - 2000 Mar

Department of Applied Physics and Physico-Informatics, Yokohama, Japan.

Supervisor: Prof. Yutaka Tomita, Prof. Takeshi Aihara, and Prof. Minoru Tsukada

Dissertation title: Identification of characteristic curve of synaptic modification
by spike timing: approach with optical recording method.



Research and Teaching Experience

Kyoto University Project-specific Associate Professor	2017 Apr - Present
Honda Research Institute Japan Co., Ltd. Senior Scientist	2016 Aug - Present
International Christian University Part-time lecturer	2014 Dec – 2017 Mar
RIKEN Brain Science Institute, Research Scientist Postdoctoral researcher at Taro Toyoizumi lab	2011 Apr - 2016 Jul
Massachusetts Institute of Technology, Visiting Researcher Postdoctoral researcher at Emery N. Brown lab	2009 Dec - 2011 Mar
RIKEN Brain Science Institute, Visiting Researcher Postdoctoral researcher at Sonja Gruen lab	2007 Aug - 2011 Mar
JSPS* Research Fellow at RIKEN Brain Science Institute	2008 Apr - 2011 Mar
JSPS* Research Fellow at Kyoto University * Japan Society for the Promotion of Science	2006 Apr - 2008 Mar

Teaching Experience

Lectures

* Lecture slides are available at <http://www.neuralengine.org/res/lecture>.

Kyoto University, Kyoto, Japan Offer lectures on computational neuroscience in “ Computational and cognitive neuroscience ” course for graduate students, organized by Prof. Takatsune Kumada and in “ Proseminar on neuroinformatics ” organized by Prof. Yasuyuki Kamitani.	2017 Apr – Present
RIKEN Brain Science Institute, Saitama, Japan An introductory course on statistics titled “ Statistics for Neuroscience ” as a part of lecture courses of the RIKEN Brain Science Training Program for selected graduate students. The course covers classical statistical tests and multivariate analysis.	2014 Dec – Present
International Christian University, Tokyo, Japan An undergraduate course on “ Statistical Physics ” at the Department of Material Science of the International Christian University. The course provides concepts in statistical physics including laws of thermodynamics, state functions, and the canonical ensemble formalism.	2014 Dec – 2017 Mar



Society for Young Researchers on Neuroscience

2014 Dec – 2017 Mar

An invited lecturer of two-days workshop organized by Society for Young Researchers on Neuroscience (in Japan), held at Hotel Wing International Sagamihara, Kanagawa, Japan. The course titled “**Introduction to time-series modeling of neural activity data**” offered basic analysis skills on spiking neural data by point process models. The course included tutorials on generation, estimation, and validation of event data, using Julia language.

The Institute of Statistics and Mathematics, Tokyo, Japan

2016 Feb

A fee-charging extension course for businesspersons and scholars on “**Introduction to statistical analysis of point process networks**” at the Institute of Statistics and Mathematics with Prof. Shinsuke Koyama. More than 72 people participated in this course.

Graduate School of Science and Technology at Meiji University, Tokyo, Japan 2013 May

Introductory courses of Neuroscience. Course titles: “**Introduction to sensory systems, perception and neural coding studies**” and “**Introduction to neural decoding**”.

The Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

2013 Mar

Invited two-week lecture course titled “**Introduction to statistical models of neural spike train data**” at School of Cognitive Sciences, IPM. This course covered recent modeling approaches for neuroscientific data with an introduction to a point process theory, generalized linear modeling, and adaptive filtering theory.

Supervision of postdoctoral researchers

Dr. Amin Sayed Moosavi (Kyoto University)

2018 Mar – present

Dr. Amin is interested in emergence of critical phenomena from neural networks, and tries to elucidate its information theoretic implications and the underlying mechanisms.

Dr. Miguel Aguilera (University of the Basque Country)

2018 Jul – 2018 Aug

Dr. Aguilera is a visiting scholar to my lab at Kyoto University, and is working on theoretical methods to measure integrated information carried in the activities of neural networks.

Supervision of graduate students

Ms. Safura Rashid Shomali (IPM, Tehran)

2013 Dec – 2017 Sep

Co-supervision of Ph.D. thesis with Prof. Majid Nili-Ahmadabadi. Ms. Shomali is a graduate student at IPM in Tehran, Iran. I met her when I gave a lecture course at her institute in March 3-12, 2013. Later she asked for my supervision, and I was officially appointed as a co-supervisor for her PhD thesis with Prof. Majid Nili-Ahmadabadi on Dec 4, 2013. She visited RIKEN during Jul 23- Sep 23 in 2014. I supervised her project on theoretical analysis of the impact of a presynaptic input on a leaky integrate-and-fire neuron receiving noisy balanced inputs. On Sep 2017, she received PhD with this work.

Mr. Christian Donner (BCCN Berlin)

2014 Jun – 2015 Apr

Supervisor of a summer-intern project at RIKEN Brain Science Institute and co-supervisor of his master thesis at Bernstein Center for Computational Neuroscience (BCCN) Berlin with Prof. Klaus Obermayer. During 5/26-8/29 in 2014, Mr. Donner stayed at RIKEN BSI to study the



dynamic model of neural interactions under my supervision, as one of three lab-rotation projects required for his Master's degree at BCCN Berlin. During his stay, he introduced an analytical approximation method to make the model applicable to large-scale analysis. I co-supervised his Master thesis on this topic with Prof. Klaus Obermayer.

Supervision of undergraduate students (internship students)

Bingyue Zhu (University of British Columbia, Canada) 2018 Jun 20 – 2019 Apr 26

Sequential Bayes estimation and anomaly detection method for physiological time-series signals

Magalie Tatischeff (University of British Columbia, Canada) 2018 Jun 20 – 2019 Apr 26

Data analysis and development of statistical models for spiking activity of neural populations

Shukla Shashwat (IIT Bombay, India) 2018 May 14 – 2018 Jul 13

Investigation of advanced time-series models for neural and behavioral activities

Krishna Subramani (IIT Bombay, India) 2018 May 14 – 2018 Jul 13

Learning a hierarchical model for spatial phases in natural scenes.

Brinti Datta (IIT Bombay, India) 2017 Jul 5 – 2017 Jul 14

Filtering methods for time-series analysis

Jimmy Gaudreault (Polytechnique Montreal) 2017 May 15 – 2018 Apr 27

Data analysis of neural population activities by the dynamic Ising model

Arunabh Saxena (IIT Bombay, India) 2017 May 8 – 2017 July 14

Studying and extending the state-space analysis methods for neural populations

Publications

Preprint

Shimazaki H. The principles of adaptation in organisms and machines I: machine learning, information theory, and thermodynamics. (2019) [arXiv:1902.11233](https://arxiv.org/abs/1902.11233)

Torigoe M, Islam T, Kakinuma H, Fung CCA, Isomura T, **Shimazaki H.**, Aoki T, Fukai T, Okamoto H. Future state prediction errors guide active avoidance behavior by adult zebrafish. (2019) doi: <https://doi.org/10.1101/546440>



Shomali SR, Ahmadabadi MN, Rasuli SN, **Shimazaki H**. Uncovering network architecture using an exact statistical input-output relation of a neuron model. *bioRxiv*. (2018)
doi: <https://doi.org/10.1101/479956>

MaBouDi H, **Shimazaki H**, Soltanian-Zadeh H, Amari S. Learning Complex Representations from Spatial Phase Statistics of Natural Scenes. *bioRxiv*. (2017)
doi: <https://doi.org/10.1101/112813>

Journal

Shomali SR, Ahmadabadi MN, **Shimazaki H**, Rasuli SN. How does transient signaling input affect the spike timing of postsynaptic neuron near the threshold regime: an analytical study. *Journal of Computational Neuroscience* (2018) 44(2), 147-171

Kass RE, Amari S, Arai K, Diekmann CO, Diesmann M, Doiron B, Fairhall A, Fiddyment GM, Fukai T, Grün S, Harrison MT, Helias M, Nakahara H, Teramae J, Thomas PJ, Reimers M, Rodu J, Rotstein HG, Shea-Brown E, **Shimazaki H**, Shinomoto S, Yu BM, Kramer MA. Computational Neuroscience: Mathematical and Statistical Perspectives. *Annual Review of Statistics and Its Application*. (2018) 5:8.1–8.32

MaBouDi H, **Shimazaki H**, Giurfa M, Chittka L. Olfactory learning without the mushroom bodies: Spiking neural network models of the honeybee lateral antennal lobe tract reveal its capacities in odour memory tasks of varied complexities. *PLoS Computational Biology* (2017) 13(6): e1005551.

Donner C, Obermeyer K, **Shimazaki H**. Approximate Inference for Time-varying Interactions and Macroscopic Dynamics of Neural Populations. *PLoS Computational Biology* (2017) 13(1): e1005309

Mochizuki Y, Onaga T, **Shimazaki H**, Shimokawa T, Tsubo Y, Kimura R, Saiki A, Sakai Y, Isomura Y, Fujisawa S, Shibata K, Hirai D, Furuta T, Kaneko T, Takahashi S, Nakazono T, Ishino S, Sakurai Y, Kitsukawa T, Lee JW, Lee H, Jung MW, Babul C, Maldonado PE, Takahashi K, Arce-McShane FI, Ross CF, Sessle BJ, Hatsopoulos NG, Brochier T, Riehle A, Chorley P, Gruen S, Nishijo H, Ichihara-Takeda S, Funahashi S, Shima K, Mushiake H, Yamane Y, Tamura H, Fujita I, Inaba N, Kawano K, Kurkin S, Fukushima K, Kurata K, Taira M, Tsutsui K, Ogawa T, Komatsu H, Koida K, Toyama K, Richmond BJ, and Shinomoto S. Similarity in Neuronal Firing Regimes across Mammalian Species. *Journal of Neuroscience* 36:5736-5747, 2016.

Chou MY, Amo R, Kinoshita M, Cherng BW, **Shimazaki H**, Agetsuma M, Shiraki T, Aoki T, Takahoko M, Yamazaki M, Higashijima S, and Okamoto H. Social conflict resolution regulated by two dorsal habenular subregions in zebrafish. *Science* 352(6281) 87-90, 2016

MaBouDi H, **Shimazaki H**, Amari S, Soltanian-Zadeh H, Representation of higher-order statistical structures in natural scenes via spatial phase distributions. *Vision Research*, Vol. 120, 61-73, 2016



Shimazaki H, Sadeghi K, Ishikawa T, Ikegaya Y, Toyoizumi T. Simultaneous silence organizes structured higher-order interactions in neural populations. *Scientific Reports*, 5, 9821, 2015.

Shimazaki H, Amari S., Brown E. N., and Gruen S., State-space analysis of time-varying higher-order spike correlation for multiple neural spike train data. *PLOS Computational Biology*, 8(3): e1002385, 2012.

Shimazaki H and Shinomoto S. Kernel bandwidth optimization in spike rate estimation. *Journal of Computational Neuroscience*, 29 (1-2) 171-182, 2010.

Shimazaki H and Shinomoto S. A method for selecting the bin size of a time histogram. *Neural Computation*, Vol. 19(6), 1503-1527, 2007.

Shimazaki H and Niebur E. Phase transitions in multiplicative competitive processes. *Physical Review E*, 72(1), 011912, 2005.

Tsukada M, Aihara T, Kobayashi Y, and **Shimazaki H**. Spatial analysis of spike-timing-dependent LTP and LTD in the CA1 area of hippocampal slices using optical imaging. *Hippocampus*, 15(1), 104-109, 2005.

(In Japanese) Kobayashi Y, **Shimazaki H**, Aihara T, Tsukada M. Spatial distributions of hippocampal LTP/LTD induced by electrically from schaffer collaterals and stratum oriens with relative timing. *The brain and neural networks*, 8 (2), 57-64, 2001.

Refereed conference proceedings

Gaudreault J, Saxena A, **Shimazaki H**. Online estimation of multiple dynamic graphs in pattern sequences. (2019) arXiv:1901.07298 accepted for IJCANN2019

Gaudreault J and **Shimazaki H**. State-space analysis of an Ising model reveals contributions of pairwise interactions to sparseness, fluctuation, and stimulus coding of monkey V1 neurons. ICANN2018, *Lecture Notes in Computer Science* 11141, 641–651, 2018

Donner C and **Shimazaki H**. Approximate inference method for dynamic interactions in larger neural populations. ICONIP 2016, *Lecture Notes in Computer Science* 9949, 104–110, 2016

Shimazaki H Single-trial estimation of stimulus and spike-history effects on time-varying ensemble spiking activity of multiple neurons: a simulation study. *Journal of Physics: Conference Series*, 473 012009, 2013.

Shimazaki H, Amari S, Brown EN, and Gruen S. State-space analysis on time-varying correlations in parallel spike sequences. *Proc. IEEE ICASSP*, 3501-3504, 2009.

Shimazaki H and Shinomoto S. A recipe for optimizing a time-histogram. *Advances in Neural Information Processing Systems*, Vol. 19, 1289-1296, 2007.



Shimazaki H and Niebur E. Correlated multiplicative modulation in coupled oscillator systems: a model of selective attention. *Progress of Theoretical Physics Supplement*, No.161 336-339, 2006.

Review papers

(In Japanese) **Shimazaki H.** The principles of adaptation in recognition and behavior. *The Brain & neural networks* (2018) 25(3) 86-103

(In Japanese) **Shimazaki H.** Analysis of multiple neural spike train data using the log-linear model: From stationary to time-varying spike correlation. *The Brain & neural networks*, 8(4) 194-203, 2011

Book Chapter

Shimazaki H. Neural Engine Hypothesis. p267-291 In: Chen Z., Sarma S. (eds) *Dynamic Neuroscience: Statistics, Modeling, and Control*. (2018) Springer, Cham (This is an extended manuscript of "Neurons as an Information-theoretic Engine" *arXiv*:1512.07855, 2015)

Shimazaki H. Japanese translation of **Principles of Neural Science 5th ed.** Kandel E.R. et al. editors, Appendix F: Theoretical Approaches to Neuroscience: Examples from Single Neurons to Networks by Abbott L.F., Fusi S., and Miller K.D., p1567-1583, Medical Science International Inc., 2014. Original: p1601-1618, McGraw-Hill, 2012.

Selected invited talks

Brain Science Workshop of Tamagawa University, Japan. Sep 28, 2017. Host Prof. Yutaka Sakai and Prof Hideo Komatsu. Title: Mathematical and statistical approaches to neural population activity

Queen Mary University of London, UK. Jul 4, 2016. Host Prof. Lars Chittka. Title: Population coding of neurons: Dynamics, higher-order interactions, and mechanisms.

Bernstein Center of Computational Neuroscience Berlin, Berlin, Germany. Apr 2015. Host Prof. Klaus Obermayer. Title: Simultaneous silence explains structured higher-order interactions of neural populations.

ELC International Meeting on "Inference, Computation, and Spin Glasses" (ICSG2013), Sapporo, Japan. Jul 2013. Title: State-space analysis of time-varying higher-order interactions: its applications to neuroscience.

The 3rd Mathematical Neuroscience Workshop in School of Mathematics, Institute for Research in Fundamental Sciences (IPM). Tehran, Iran. Mar 2013. Title: The simultaneous silence of neurons explains higher-order interactions in ensemble spiking activity.



Workshop on Neural Information Flow, Kyoto University, Kyoto, Japan. Jun 2012. Title: Tracking dynamic neural interactions in awake behaving animals.

Neurostatistics Working Group Seminar, Dept. of Biostatistics, Harvard University, Dec 2010. Title: Analysis of dynamic neural spike data: from firing rates to spike correlations.

IEEE ICASSP2009, Special Session on 'Signal Processing for Neural Spike Trains', Taipei, Taiwan. Apr 2009. Title: State-space analysis on time-varying correlations in parallel spike sequences.

NIPS 2008, Workshop on 'Statistical Analysis and Modeling of Response Dependencies in Neural Populations', Whistler, Canada. Dec 2008. Title: State-space analysis on time-varying higher-order spike correlations.

Riken Brain Science Institute Forums, Wako, Japan. Host: Dr. Sonja Gruen. Apr 2007. Title: A recipe for optimizing a time histogram of spike data.

The Boadian Seminar at Mind/Brain Institute, Johns Hopkins University. Baltimore, USA. Host: Dr. Ernst Niebur. Mar 2007. Title: A recipe for constructing a peri-stimulus time histogram.

Awards

Excellent paper award at The 23rd International Conference on Neural Information Processing (ICONIP2016)	2016 Oct
Letter of Appreciation from the RIKEN president	2009 Oct
Japanese Neural Network Society Research Award 2009	2009 Sep 26
Japanese Neural Network Society Young Researcher Award 2007	2007 Nov 15

Research Grants and Fellowships

Excellent Young Researchers Overseas Visit Program by JSPS	2009 Dec-2011 Mar
Research Fellowship for Young Scientists by JSPS PD	2008 Apr-2011 Mar
Research Fellowship for Young Scientists by JSPS DC	2006 Apr-2008 Mar
Japan Student Services Organization (Fellowship)	2004-2006
Solo Snyder Foundation, Johns Hopkins University (Fellowship)	2002-2004
Murata Overseas Scholarship (Fellowship and Research Grant)	2000-2002

Travel Award



The 21st Century COE Center for Diversity and Universality in Physics. 2006 Dec

Tamura Memorial Foundation

2006 May

Japanese Neural Network Society

2006 Jan

Academic service

A board member, Japanese Neural Network Society

2018 Mar - present

List of all presentations

■ 2019

Shimazaki H. Network architecture underlying sparse neural activity characterized by structured higher-order interactions. Okinawa Institute of Technology (OIST). Okinawa, Japan. Mar 26 2019 **TALK**

Moosavi S.A., Shimazaki H. Role of gain-control and neural correlations in efficient stimulus coding. Consciousness Research Network (CoRN 2019) Okazaki, Japan. Feb 23 2019 **POSTER**

Tatischeff M., Gaudreault J., Donner C., Shimazaki H.. Thermodynamic analysis of neural populations by the state-space Ising model. Consciousness Research Network (CoRN 2019). Okazaki, Japan. Feb 23 2019 **POSTER**

■ 2018

Shimazaki H., Tatischeff M., Gaudreault J., Donner C. Thermodynamic analyses of neural populations. Analysis and Synthesis for Human/Artificial Cognition and Behaviour, A satellite workshop of JNNS2018. Okinawa Institute of Science and Technology Graduate University. Oct 22, 2018 **POSTER**

Gaudreault J. and Shimazaki H. State-space analysis of an Ising model reveals contributions of pairwise interactions to sparseness, fluctuation, and stimulus coding of monkey V1 neurons. The 27th International Conference on Artificial Neural Networks (ICANN2018) Oct 5, 2018 **POSTER**

Shimazaki H. Neural network architecture that produces sparse population activity. Workshop on multitrack event-trains in neural, social, seismological, and financial data. Karuizawa, Japan. July 7, 2018. **TALK** (In Japanese)

Gaudreault J, Saxena A, Shimazaki H. State-space analysis of an Ising model reveals contributions of pairwise interactions to sparseness, fluctuation, and stimulus coding of monkey V1 neurons. Mechanism of Brain and Mind, Rusutsu, Hokkaido, Japan. Jan 9, 2018 **POSTER**

■ 2017



Shimazaki H. Network architectures underlying variable, sparse population activity of neurons. Fluctuations of event occurrences in a variety of networks. Kyoto, Japan. Nov 15, 2017 **TALK**

Shimazaki H. Mathematical and statistical approaches to neural population activity. Brain Science Workshop of Tamagawa University. Isawa-onsen, Yamanashi, Japan. Sep 28, 2017. **INVITED TALK** (In Japanese)

Makio Torigoe, Islam Tanvir, Hisaya Kakinuma, Hideaki Shimazaki, Chi Chung Alan Fung, Tazu Aoki, Tomoki Fukai, Hitoshi Okamoto In-vivo imaging of telencephalic neural activities in adult zebrafish performing decision making task in the closed-loop virtual reality environment. Annual Meeting of the Japan Neuroscience Society (Neuro2017). Makuhari Messe, Japan. Jul 23 2017. **INVITED TALK**

■ 2016

Donner C. and Shimazaki H., Estimating dynamic functional networks of larger neural populations. Society for Neuroscience 2016. Nov 16, 2016

Shomali S.R., Ahmadabadi M.N., Rasuli N. S., Shimazaki H., Exact analysis of spike-timing and higher-order interactions of neurons at the threshold regime suggests network architecture underlying sparse population activity. Society for Neuroscience 2016. Nov 14, 2016

Donner C. and Shimazaki H., Large-scale inference of time-varying neural interactions. The 23rd International Conference on Neural Information Processing (ICONIP 2016). Kyoto, Japan. Oct 16, 2016 (Talk by Donner) **EXCELLENT PAPER AWARD**

MaBouDi H., Shimazaki H., Lars Chittka L. Modelling elemental learning of honeybees by spiking neural networks. EURBEE 2016. Cluj-Napoca, Romania. Sep 7, 2016 (Talk by MaBouDi)

Shimazaki H. Simultaneous silence explains structured higher-order interactions of neural populations. Modeling Neural Activity: Statistics, Dynamical Systems, and Networks, Waikoloa, Hawaii, USA. Jun 22-24 2016, **TALK**

Shimazaki H. Toward thermodynamic principles of consciousness. University of Sussex, UK. Jul 7, 2016. **INVITED TALK**

Shimazaki H. Higher-order interactions of neural populations. University College London, UK. Jul 5, 2016. **TALK**

Shimazaki H. Population coding of neurons: Dynamics, higher-order interactions, and mechanisms. Queen Mary University of London, UK. Jul 4, 2016. **INVITED TALK**

Shomali S.R., Ahmadabad M.N., Shimazaki H., Rasuli SN. Exact spike-timing distribution reveals higher-order interactions. CNS*2016. Jeju, South Korea. Jul 2, 2016 (Poster by Shomali) **POSTER (Reviewed)**

Shimazaki H. Analysis of network activity of neurons by the dynamic Ising model. International Christian University, Tokyo, Japan. Feb 2, 2016 **INVITED TALK**



Shimazaki H. Simultaneous silence explains structured higher-order interactions of neural population. MONA2 - Modelling Neural Activity. Hawaii, USA. Jun 22, 2016 **TALK**

Shimazaki H., Peters A.J., Komiyama T., Toyozumi T. Redundant coding by layer 2/3 neurons of motor cortex during initial motor learning. Mechanism of Brain and Mind, Rusutsu, Hokkaido, Japan. Jan 6-8, 2016 **POSTER**

■ 2015

Donner C. and Shimazaki H., Approximation methods for inferring time-varying interactions of a large neural population. Workshop in Statistical Methods for Understanding Neural Systems at Neural Information Processing Systems (NIPS) 2015. Montreal Canada, Dec 11, 2015 **POSTER (Reviewed)**

Shimazaki H. Simultaneous silence explains structured higher-order interactions of neural populations. Juelich Research Center. Juelich, Germany. May 5, 2015 **INVITED TALK**

Shimazaki H. Simultaneous silence explains structured higher-order interactions of neural populations. Bernstein Center of Computational Neuroscience Berlin. Berlin, Germany. Apr 28, 2015 **TALK**

■ 2014

Sharp T., Shimazaki H., Isomura Y., Fukai T. State-space analysis of behaviour- and layer-dependent synchrony in motor cortex during volitional arm movement. Society for Neuroscience (SfN) 2014, Washinton DC, USA. Nov 16, 2014. **POSTER**

Sharp T., Shimazaki H., Isomura Y., Fukai T. Behaviour- and layer-dependent synchrony in motor cortex during volitional arm movement. Neuro2014, Yokohama, Japan. Sep 11, 2014. **POSTER**

Shomali S.R., Ahmadabadi M.N., Shimazaki H., Rasuli N. S. Theoretical study on spike-timing probability in a pair of pre-post synaptic neurons. Neuro2014, Yokohama, Japan. Sep 11, 2014. **POSTER**

MaBouDi H., Shimazaki H., Abouzari M., Soltanian-Zadeh H., Amari S. Bimodal distributions of local phase variables in natural images. The 2014 Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, Florida, USA. May 18, 2014 **POSTER (Reviewed)**

Sharp T., Shimazaki H., Isomura Y., Fukai T. Behaviour- and layer-dependent synchrony in motor cortex during volitional arm movement. Workshop on data mining in neuroscience. National Institute of Informatics, Tokyo, Japan. May 28-29, 2014 **TALK**

MaBouDi H., Shimazaki H., Abouzari M., Amari S, Soltanian-Zadeh H. Statistical inference for directed phase coupling in neural oscillators. Computational and Systems Neuroscience (Cosyne) 2014. Salt Lake City, USA. Feb 27 **POSTER (Reviewed)**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T. Structured higher-order interactions explain the simultaneous silence of neural populations. Mechanism of Brain and Mind, Rusutsu, Hokkaido, Japan. Jan 8-10, 2014 **POSTER**

■ 2013



Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T., The simultaneous silence of neurons explains structured higher-order interactions in ensemble spiking activity. Society for Neuroscience (SfN). San Diego, USA. Nov 9-13 **POSTER**

Shimazaki H., State-space Analysis of Time-varying Higher-order Interactions: its Applications to Neuroscience. ELC International Meeting on "Inference, Computation, and Spin Glasses" (ICSG2013). Sapporo, Japan. Jul 28 **INVITED TALK**

Shimazaki H., Higher-order interactions in population activity of hippocampal CA3 neurons. Workshop on statistical analysis of neurophysiological and clinical data, Kyoto, Japan. Jul 8-9 **TALK**

Shimazaki H., Estimating Time-varying Higher-order Neuronal Interactions in Awake Behaving Animals. Modeling Neural Activity: Statistics, Dynamical Systems, and Networks, Lihue, Hawaii, USA. 2013 Jun 26-28 **TALK**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T., The simultaneous silence of neurons explains higher-order interactions in ensemble spiking activity. Neuro2013, Kyoto, Japan. Jun 20 **POSTER**

Shimazaki H. Estimating dynamic neural interactions in awake behaving animals. The 3rd Mathematical Neuroscience Workshop in School of Mathematics, Institute for Research in Fundamental Sciences (IPM). Tehran, Iran. Mar 13 **INVITED TALK**

Shimazaki H. The Simultaneous Silence of Neurons Explains Higher-Order Interactions in Ensemble Spiking Activity. The 3rd Mathematical Neuroscience Workshop in School of Mathematics, Institute for Research in Fundamental Sciences (IPM). Tehran, Iran. Mar 14 **INVITED TALK**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T. The simultaneous silence of neurons explains structured higher-order interactions in spontaneous spiking activity. Dialog between Neuroscience and Statistics, Institute of Mathematics and Statistics, Tokyo, Japan. Feb 18-19 **TALK**

■ 2012

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T. Joint inactivation statistics of population spiking activities. RIKEN BSI Retreat 2012, Karuizawa, Japan. Nov 12-13 **POSTER**

Shimazaki H. The simultaneous silence of neurons explains structured higher-order interactions in ensemble spiking activity. RIKEN BSI Lunch Seminar 2012, Wako, Japan. Nov 8 **TALK**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T. Joint inactivation statistics of population spiking activities. Workshop on statistical aspects of neural coding. Nov 1-2. Kyoto University & Ritsumeikan University. **TALK**

Shimazaki H. Tracking Dynamic Neural Interactions in Awake Behaving Animals. Workshop on neural information flow, Kyoto University, Kyoto, Japan. Jun 20 **INVITED TALK**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T. The simultaneous silence of neurons explains higher-order interactions in ensemble spiking activity. Computational and Systems Neuroscience (Cosyne) 2012. Salt Lake City, USA. Feb 23-26 **POSTER (Reviewed)**



■ 2011

Shimazaki H., Amari S., Brown E. N., and Gruen S. Dynamics of Higher-order Spike Correlation in an Awake Behaving Monkey: Analysis by a State-space Model. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

Shimazaki H., Ikegaya Y., and Toyozumi T. A New Sparse Log-linear Model for Simultaneously Active and Inactive Neurons. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

Shimazaki H. and Brown E.N. Copula-based Mixture Time-series Model of Continuous and Point Processes for Synthetic Analysis of Neural Signals. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

Shimazaki H. and Brown E. N. Constructing a joint time-series model of continuous and Bernoulli/Poisson processes using a copula. Computational and Systems Neuroscience (Cosyne) 2011, Salt Lake City, USA. Feb 24-27 **POSTER (Reviewed)**

■ 2010

Shimazaki H. Analysis of Dynamic Neural Spike Data: From Firing Rates to Spike Correlations. Neurostatistics Working Group Seminar, Dept. of Biostatistics, Harvard University, Boston, USA. Dec 1 **INVITED TALK**

Shimazaki H. Detection of dynamic cell assemblies by the Bayes Factor. Workshop on spatio-temporal neuronal computation, Kyoto University, Kyoto, Japan. Sep 6-7 **INVITED TALK**

Shinomoto S*, Shimazaki H, and Shimokawa T. Characterizing neuronal firing with the rate and the irregularity. Neuro2010, Kobe, Japan. Sep 2 S3-10-1-3 **TALK**

Shimazaki H., Gruen S., and Amari S. Analysis of subsets of higher-order correlated neurons based on marginal correlation coordinates. Computational and Systems Neuroscience (Cosyne) 2010, Salt Lake City, USA. Feb 25-28 **POSTER (Reviewed)**

■ 2009

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Model of Dynamic Spike Correlation. Japanese Neural Network Society 2009. Sendai, Japan. Sept. 24-26. TALK+POSTER **Japanese Neural Network Society 2009 Distinguished Research Award**

Shimazaki H. Analysis of Dynamic Spike Data: From Spike Rate to Multiple Neuron Spike Correlation. The 6th Brain Lunch Seminar. RIKEN Brain Science Institute, Saitama, Japan. Sep 8. **TALK**

Shimazaki H., Amari S., Brown E. N., and Gruen S. Bayes Factor Analysis for Detection of Time-dependent Higher-order Spike Correlations. CNS2009. Berlin, Germany. Jul 18-23 P99 **POSTER**

Shimazaki H. and Shinomoto S. Histogram binwidth and kernel bandwidth selection for the Spike-rate estimation. CNS2009. Berlin, Germany. Jul 18-23 P116 **POSTER**



Shimazaki H., Amari S., Brown E. N., and Gruen S. Estimating time-varying spike correlations from parallel spike sequences, German-Japanese Workshop "Computational and Systems Neuroscience", Berlin, Germany. May 25-28 **POSTER**

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Analysis on Time-varying Correlations in Parallel Spike Sequences. IEEE ICASSP2009 Special Session on 'Signal Processing for Neural Spike Trains', Taipei, Taiwan. Apr 24 SS-L10.4 **INVITED LECTURE**

Shimazaki H., Amari S., Brown E. N., and Gruen S. Detection of non-stationary higher-order spike correlation. Cosyne 2009, Salt Lake City, USA. Feb 26 -Mar 3 II-62 **POSTER (Reviewed)**

■ 2008

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